
PHYSICAL
ACOUSTICS

Sound Generation as a Drop Falls on a Water Surface

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Received March 12, 2011

Abstract—Impact of a drop on a water surface is accompanied by a series of sound pulses propagating in air and underwater. Depending on the falling mode (drop size and initial velocity), pulses substantially differ in amplitude, duration, and modulation frequency. We study falling modes in which in addition to conventional sound packets—the shock pulse and single resonance sound packets—several packets are observed. Experiments were conducted with simultaneous recording of sound in air and underwater and were accompanied by synchronous video depiction of currents in the drop impact region. Comparison of videograms and phonograms demonstrate that the sources of sound packets are gas cavities of arbitrary shape detached from the underwater cavern under the action of large accelerations (several km/s^2) during a sharp change in its surface area, which gradually achieve equilibrial elliptical and spherical shapes.

Keywords: drop, underlying surface, collision, cavern, cavity, acoustic emission, shock pulse, resonance, sound packet, microphone, hydrophone, recording.

DOI: 10.1134/S1063771011050137

The study of processes accompanying collision of a vertically falling drop with a motionless water surface—evolution of the cavern and the upward stream and the cascade of vortices propagating in the bulk of a fluid—which were begun back in the 19th century [1], continue to attract the attention of physicists owing to their abundance in nature and importance of practical applications, particularly acoustic ones [2]. Splashes intensify matter exchange between the hydro- and atmosphere. Secondary small drops expelled from the underlying fluid are carried off by the wind and, evaporating, leave microcrystals of dissolved substances and other hard fractions in the atmosphere, which act as water vapor condensation centers. Microdrops can also carry harmful substances and microorganisms into the atmosphere.

The falling of a drop is accompanied by distortion in the shape of the free surface, which is easily detected by remote probing methods in radiowave ranges and serves as a source of valuable information on the energetics and structures of processes at the ocean–atmosphere contact surface [3]. The data obtained are used in estimating near-surface wind velocity and precipitation intensity in remote regions of the world ocean that have no meteorological stations [2, 3, 5].

Systematic recording of sound accompanying the falling of drops in fluid began in the 1920s, first in the atmosphere [6], and then with the appearance of HF hydrophones, in the hydrosphere. A simple formula for the frequency of sound emitted in water by an oscillating bubble, obtained in [7], is still used for estimates today.

The hydroacoustics of splashes of individual drops and rain began intensive development during World War II, when problems of concealment and remote detection of submarines assumed particular importance. An apparatus that had been developed for military purposes began to be used for meteorology in peacetime [8], in particular, to estimate precipitation intensity in remote water areas. At the first stage, the technical possibilities of the apparatus made it possible to find only qualitative regularities, in particular, characterizing the relationship between the spectral characteristics of underwater sound and precipitation intensity [9].

Improvement in the apparatus—an increase in sensitivity and temporal and spatial resolution—made it possible to begin studying sound emitted by an individual drop. The widely cited work [10] proposed several sound emission mechanisms: the mechanism owing to hydraulic impact upon primary contact of the drop with the fluid surface, and the resonance mechanism, owing to excitation of gas bubble vibrations observed in splash patterns. The horn emission mechanism was also discussed, owing to resonance oscillations of the walls of the cavern that forms in the fluid after drop absorption. Subsequent studies demonstrated that in the records of underwater acoustic signals obtained during rain, a spectral peak is steadily pronounced in the range of 10–15 kHz [10]. From the results of experiments it was established that the impact and resonance mechanisms participate in sound emission; the influence of the horn mechanism has not been confirmed [11].

In the interest of elaborating sound emission processes, development of an apparatus was begun in which instruments for high-speed filming of the flow pattern were synchronized with devices for measuring sound pressure in water [12]. From the results of experiments, a diagram of flow modes was obtained on the falling velocity–drop diameter plane, in which the region of the stable existence of emitting gas bubbles was isolated for the first time, explaining the presence of the spectral peak [10] with allowance for the relationship between the resonance oscillation frequency and bubble dimensions [7].

The conditions for the involvement of gas bubbles during drop impact and the laws of the emission intensity decrease with distance were studied in detail in [13]. The angular distribution of the emission level in the horizontal plane indicates that the sound source can be represented as a vertical dipole radiator whose moment is a function of frequency.

In the theoretical model of sound emission, at the stage of initial contact of the drop owing to supersonic traveling of the drop–underlying surface contact line, estimates were obtained on the emitted energy $E \sim \frac{3}{16} M^3 E_0$, where $E_0 = \frac{4}{6} \rho_0 U_d^2 \pi R_d^3$ is the kinetic energy of a drop with radius R_d falling at velocity U_d and M is the Mach number of the moving contact line [13]. Evaluations of pressure jumps given by this model substantially differ from estimates derived from the hydraulic impact model, in which the pressure jump $\Delta P = \rho U_d c$, where ρ and c are density and sound velocity in the underlying fluid.

Experiments revealed the standard sequence of sound signals containing the shock pulse and the main sound packet following with a large delay (up to 0.25 s). These data made it possible to construct practically important empirical regularities relating the precipitation intensity to the sound pulse spectral parameters [14].

Subsequent experiments demonstrated that sound is emitted not only by water drops, but also by slowly falling snowflakes [15]. Like in the case of drops, sound sources were gas bubbles contained in the ice crystals of snowflakes.

The acoustic measurements of individual drops, as well as artificial and natural rain, begun in [11, 12, 14] are still being actively carried out today [16]. However, the achieved rate of spatial resolution of optic images and temporal resolution of acoustic signals have proved insufficient to precisely link up short sound pulse emission processes and the rapidly changeable flow pattern.

The aim of this work is to study the mutual relations between hydrophysical and acoustic processes accompanying a drop falling in a liquid using a high-resolution system for synchronous video recording of the flow pattern and acoustic signals. The dynamic parameters of the system make it possible to link the

peculiarities in the flow pattern to the characteristics of sound signals propagating both in water and in air.

BASIC PARAMETERS OF THE PROCESS OF A DROP COLLISION WITH AN UNDERLYING FLUID

A drop of fluid (volume V_d and diameter D) with mass m , density ρ_d , kinematic viscosity ν_d , and surface tension coefficient σ_d falls from a height H_d . At the moment of contact, the drop has a velocity U_d , momentum $p_d = mU_d$, and kinetic energy $E_d = mU_d^2/2$. During detachment, volumetric vibrations are excited in the drop, the physics of which in the ideal fluid approach was examined in [17].

The drop falls freely in air (density ρ_a , kinematic viscosity ν_a , pressure P_a , temperature T_a , and gravity acceleration g) and is deformed under the action of resistance forces and the pressure gradient. The interaction of the drop with the incident flow leads to a change in its shape from spherical to bell-shaped depending on the mass and distance of fall [18]. Within the parameter range of the experiment, it has the shape of a sphere or ellipsoid of revolution.

The drop interacts with the underlying fluid (the target), which is characterized by density ρ_t , kinematic viscosity ν_t , temperature T_t , surface tension coefficient σ_t , and sound propagation velocity c .

The large number of parameters of different dimensions impedes qualitative analysis of the processes. To simplify it, the parameters are commonly brought to single dimension, usually length and time (frequency), or to dimensionless form.

The following have been distinguished from the dimensional parameters: the capillary scale $\lambda_t = \sqrt{\frac{\sigma_t}{g\rho_t}}$ and the sound wavelength $\lambda_s = c/\omega$, where ω is the sound frequency. Surface perturbations—gravitational-capillary waves—are characterized by length $\lambda_c = 2\pi/k$ (k is the wavenumber) and wave frequency ω_c , which are related by the dispersion equation $\omega_c^2 = gk(1 + k^2\lambda_t^2)$ [19].

In addition to large $t_c = 2\pi/\omega_c$, a small timescale exists in the problem, which is determined by the duration of the process of the drop merging with the fluid target, closing of open surfaces with area S into gas-filled cavities, or breaking of fluid-target surface continuity with the formation of secondary drops. As

well, within a short time $\tau_s = \left(\frac{1}{S} \left| \frac{dS}{dt} \right| \right)^{-1}$ ($\tau_s \ll 1/\omega_c$),

part of the internal energy $\varepsilon = \sigma_d S_d + \sigma_t S_t$ is rapidly converted into pressure perturbations, leading to the generation of microscale flows or, the opposite, part of the energy of large-scale mechanical motion goes to increasing the free surface. The given processes play

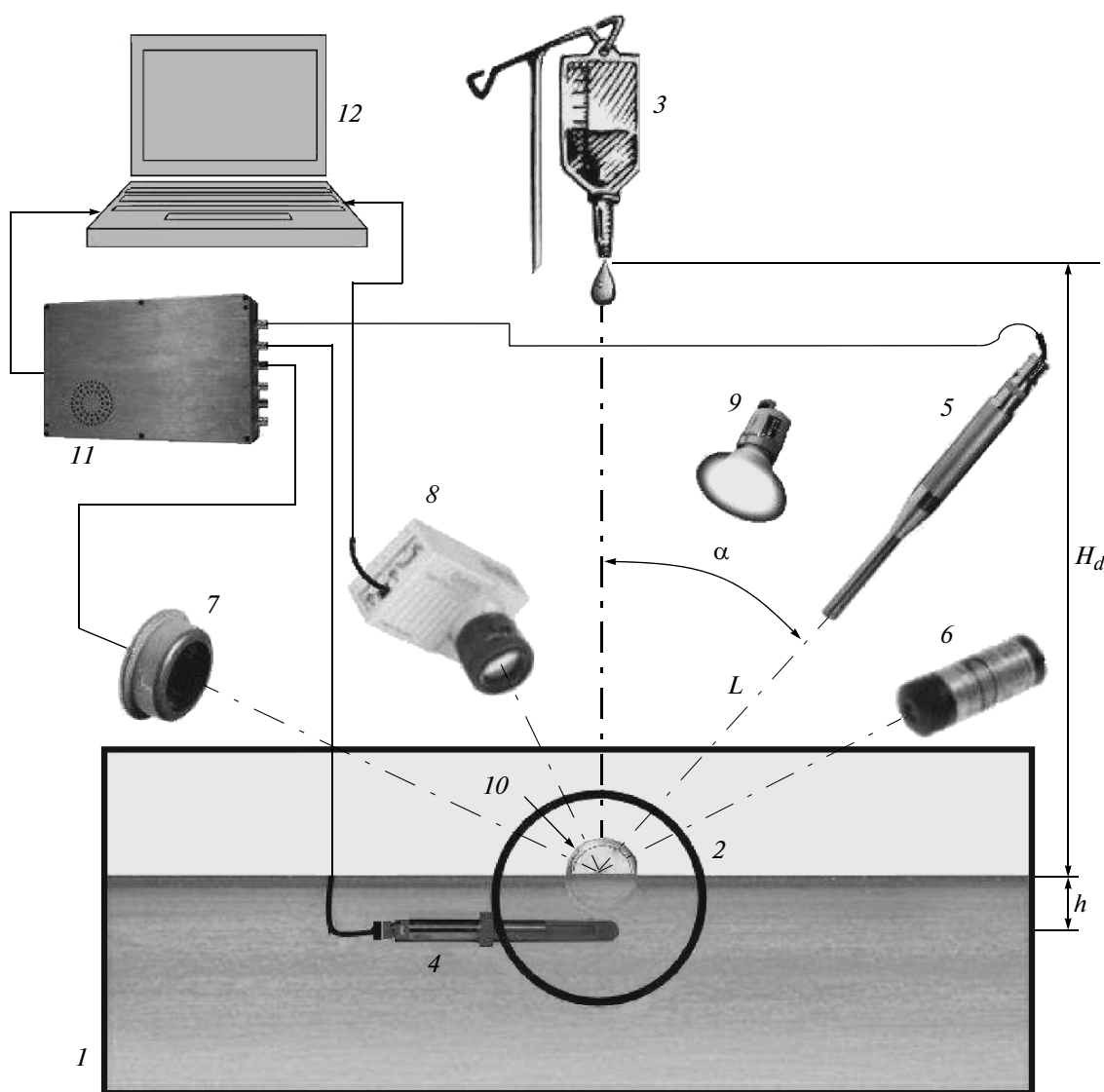


Fig. 1. Diagram of experimental setup: (1), tank; (2), viewing ports; (3), dosing unit; (4), hydrophone; (5), microphone; (6), laser; (7), photoreceiver; (8), video camera; (9), illuminator; (10), observation region; (11), interface; (12), computer.

the key role in generating acoustic and short capillary waves.

The scales determine the requirements on the spatial and temporal resolution of the recording device and on the conditions for synchronizing optic and acoustic devices.

As dimensionless relations of the problem, normally Weber number $We = \rho U_d^2 D / \sigma$, Bond number $Bo = \rho g D^2 / \sigma$, and Reynolds number $Re = U_d D / \nu$ are chosen. The Froude number is also used [2], which here is $Fr = We / Bo = U_d^2 / Dg$ (when there are material parameters, the indices are omitted, since in the given experiments the fluids of the drop and target are the same).

Under real conditions, the surface tension coefficient, density, and kinematic viscosity depend on temperature. Inhomogeneity of the surface temperature can lead to the formation of Marangoni convection, which affects the interaction character between the drop and the underlying fluid during primary contact. In this study, these effects are not considered.

EXPERIMENTAL SETUP

Figure 1 shows the diagram of the stand on which the experiments were conducted.

The main element of the setup is a hydrooptical tank 1 with dimensions of $145 \times 50 \times 60$ cm with viewing ports 2 made of optic glass. The tank was filled with degassed tapwater. The water depth in the tank

was 40 cm, which substantially exceeded all characteristic scales of hydrodynamic processes.

To obtain single drops and their sequences, a dispensing unit 3 was used, set up at height H_d above the free surface. The calibrated nozzles of the dispensing unit make it possible to obtain drops of various sizes. In these experiments, the drop diameter was 0.5 cm. The height of the dispensing unit was altered in the range from 8 to 110 cm; as well, modes were realized in which a thick cumulative stream was push out from the underlying fluid into the air. Other types of flows arising during the falling of a drop into the fluid and the experimental conditions corresponding to them are presented in [20].

A GI51B-type measuring hydrophone with a cylindrical sensing element and preamplifier was placed in the tank. The sensitive head of the hydrophone was at depth h (Fig. 1) along the line of drop fall or at a certain distance L_s from it. The sensitivity of the hydrophone was 30 mV/Pa over the 2–120 000 Hz band, and the bandpass flatness was at least 3 dB.

For sound recording in air, an RFT measuring complex with an MV-101 microphone head was applied (frequency band 20–40 000 Hz, sensitivity 1.58 V/Pa, AFC flatness as good as 3 dB). The membrane of the microphone head 5 was located at distance L from drop impact point. The pattern axis of the microphone directed at the drop impact point made an angle α with the vertical.

Perturbations at a given point of the water surface were detected by a differential optical system consisting of illuminator 6 and photoreceiver 7. The illuminator was an IE75-05PF laser diode (wavelength 0.65 μm , power 5 mW) with a collimator allowing focusing of the laser mark at the given surface point of the tank. Differential photo-detector 7 was an FD-263 photodiode (optical range 0.4–1.1 μm).

The flow pattern was recorded by an Optronis CR3000 $\times$ 2 high-speed video camera 8 (CCD sensor 13.57 $\times$ 13.68 mm, maximum frame rate 100 000 fps, 8 Gb internal memory, maximum image size 3 megapixels). The observation region was illuminated by two RayLab Xenos RH 1000 type floodlights 9 with controllable brightness. In these experiments, the frame rate was from 2000 to 20 000 fps.

The line of view of camera 8 was horizontal or at a relatively small angle to the horizontal in order to observe the perturbation pattern in the bulk of the fluid and deformation of the free surface in the drop region 10. In the diagram in Fig. 1, the camera is tuned for recording perturbations of the free surface.

A special interface block 11 was designed to control the experiment, synchronization of the data acquisition from various sensors, signals express-processing, and preparation of data for recording. The block provides 12-bit analog-to-digital conversion, and receipt and transmission of data to computer 12 for storage

and subsequent processing over four independent channels with a clock speed of up to 10 MHz.

During the measurements the video records and phonograms were synchronized with an accuracy of up to 1 μs with the help of programming block 11.

At the beginning of each experiment, the image scale at the given resolution was calibrated. The data input routine provided for both separate and simultaneous starts of video camera 8 and block 11 with controlled delay. In these experiments, the camera and block were started simultaneously to ensure synchronicity of the video picture and acoustic signals.

In contrast to the earlier experiments in which acoustic signals were recorded separately in either an air medium [7] or water medium [2], in these experiments, acoustic measurements were conducted in both media simultaneously with video recording of the flow pattern. From a wide set of flow types observed in earlier experiments with a drop falling into water at room temperature [20], only two were chosen in order to study the acoustic properties. In the selected conditions, a thick cumulative jet is expelled, either by itself or accompanied by formation of a streamer—a secondary thin jet breaking up into a chain of drops.

MAIN RESULTS

The experiments showed a noticeable variability in the acoustic signal even under invariable experimental conditions (the height of fall and the dimensions of the drops, the positioning of receiving devices); however, a number of its properties agree with those established earlier [2, 11].

The general character of the temporal dependence of signals from the hydrophone and microphone are illustrated by curves 1 and 2 in the joint phonogram (Fig. 2), respectively (the difference in grass thickness is a result of different temperature scales). A drop fell from a height $H_d = 94$ cm; velocity at the moment of impact was $U_d = 4$ m/s (from video recording data).

In the joint phonogram containing the signal of the hydrophone (1, reference) and microphone (2, offset by a delay value of $\tau = 300$ μs of the propagation signal along path L), shock pulse I and main sound packet II following with a large delay are distinguished (Fig. 2). Time t on the abscissa is counted from the leading edge of shock pulse I; along the ordinate axis, sound pressure P or normalized spectral density S are given. The high clock frequency of block 11 (1 MHz) makes it possible to trace fine signal details.

Shock pulse I with an amplitude of 9.5 Pa and duration of 140 μs arising during the collision of a drop with the underlying fluid has a steep leading edge (a steepness of 3 Pa/ μs). Further, a gradual drop and increasing pressure follow as a result of vibrations of the free surface of the fluid (a local secondary gradual maximum is achieved at $t = 53$ μs). After a long quiet period, the main sound packet II is detected.

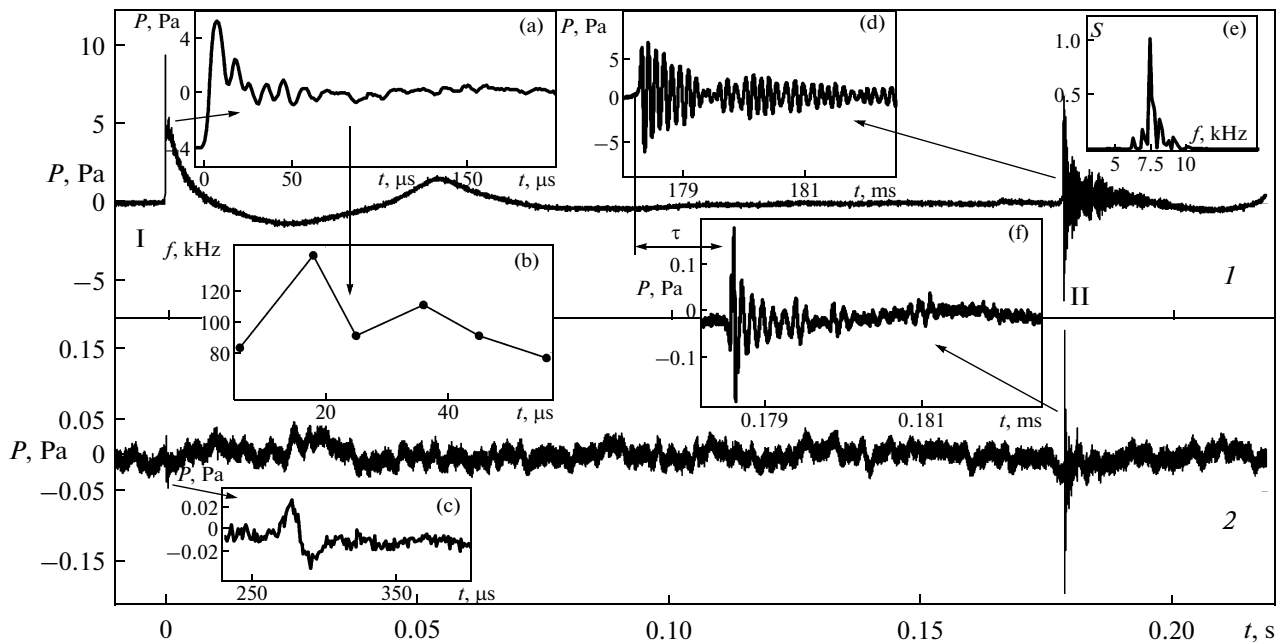


Fig. 2. Phonograms of hydrophone (I) and microphone (2) signals with sound packets in water and air. $H_d = 94$ cm, $U_d = 4$ m/s, $We = 1080$, $Bo = 3.3$, $Fr = 326$, $Re = 20000$.

In the expanded fragment of the initial signal (Fig. 2a), the increasing shock front is followed by the region of a gradual drop with superposed variable frequency oscillations, whose dependence on time is shown in Fig. 2b. Within the pulse time, the frequency decreases from 140 to 80 kHz. As a result of limitation of the microphone band, the impulse signal at its output is represented by only a single reversal pulse with the peak-to-peak amplitude of 0.04 Pa (Fig. 2c). The intense sound packet in air is detected almost simultaneously with hydrophone pulse II.

The main signal at the hydrophone output (Fig. 2d), which appears after 178 μ s, is a modulated packet with the main modulation frequency of 7.5 kHz and side frequencies in the band from 4.5 to 9 kHz (normalized spectrum in Fig. 2e).

Packet II at the microphone output (Fig. 2f) has the same central modulation frequency, but a more diffuse spectrum as a result of the lower signal-to-noise ratio. Uncontrolled LF grass oscillations reflect the acoustic noise level of the laboratory.

In all the experiments, the shock pulse was certainly recorded, whereas the parameters of the other signals noticeably changed even with preserved experimental conditions and devices. Figure 3 shows a characteristic example in which the main packet was nearly absent altogether under a starkly pronounced initial pulse. The experiment was conducted under the same conditions as the previous one. The signal at the hydrophone output includes the initial pulse and the drop/rise with a maximum at $t = 55$ ms and a weak sound pulse at $t = 176$ ms.

The initial pulse levels from the hydrophone in Figs. 2 and 3 are approximately identical; however, in the second case, the signal appears more complex (Figs. 2a, 3a). Oscillations in the decreasing area adjacent to the leading edge (up to 95 μ s) have a complex spectrum in which two peaks are pronounced at 80 and 120 kHz (normalized spectrum in Figs. 2b, 3b). They are changed by vibrations of substantially lower frequencies $f_1 = 40$ kHz and $f_2 = 35$ kHz, after which a new HF packet follows ($f_3 = 86$ kHz); further, with a delay, oscillations with the lowest frequency of $f_4 = 24$ kHz appear.

The main packet II following with a delay of $\Delta t = 176$ ms represents a single video pulse with an amplitude of 0.15 Pa (Fig. 3c), significantly less than the amplitude of the shock pulse (8 Pa). Such a composition of the sound signal is atypical for the majority of experiments in which there was one or more main packets.

In the phonogram of microphone 2, there is only one shock pulse, whose enlarged plot is shown in Fig. 3d, where τ is the delay relative to the leading edge of the shock pulse of the hydrophone. Following the weak sign-changing pulse at the microphone output, there is a sequence of signals also presented in the extended plot of the hydrophone phonogram (Fig. 3a) with the exception of the signal with frequency f_3 which is out of the microphone band pass. There is no main packet in the microphone signal.

It is possible to infer about the position of the emission source according to delay τ between the arrival of the signals to the hydrophone and microphone. The path along which sound propagates can be represented

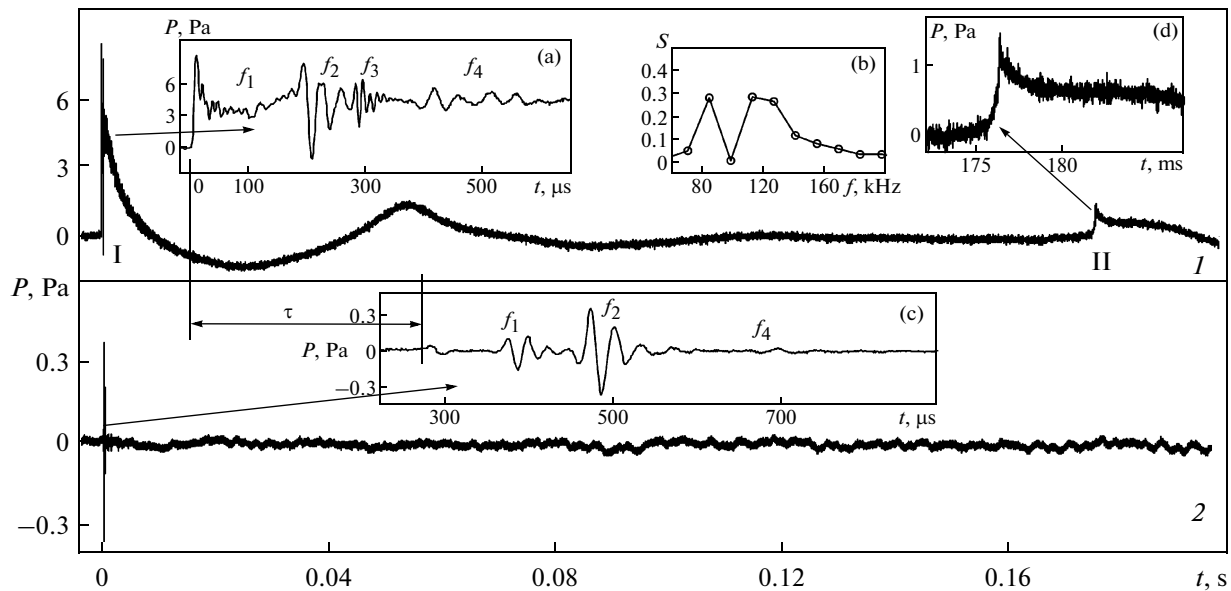


Fig. 3. Phonograms of hydrophone (1) and microphone (2) with a single sound packet in water. $H_d = 94$ cm, $U_d = 4$ m/s, $We = 1080$, $Bo = 3.3$, $Fr = 326$, $Re = 20000$.

as three components— L , L_h , and L_m , where L_h and L_m are underwater segments of the path from the sound source to the hydrophone and in the direction of the microphone; at that, the delay is $\tau = \tau_a - \frac{L_h - L_m}{c_w}$,

where $\tau_a = \frac{L}{c_a}$ is the sound propagation time for air.

With the given geometry of the experiment, the minimum value $\tau_{\min} = \tau_a - \frac{L_H}{c_w}$ (L_H is the distance from the hydrophone to the point where a drop falls) will be in the case when the sound source is located at the surface, i.e., when $L_m = 0$. For underwater positioning, the options $\tau < \tau_a$ or $\tau > \tau_a$ are possible, corresponding to cases when the source is located closer to the surface ($L_h > L_m$) or to the hydrophone ($L_h < L_m$).

For the signals in Fig. 3, $\tau_a = 295$ μ s, $\tau_{\min} = 250$ μ s, and the measured value of the delay $\tau = 280$ μ s, which points to an underwater location of the sound source closer to the hydrophone. Note that in the previous experiment (Fig. 2), which was conducted with the same values of $\tau_{\min}$ and τ_a , the value of delay $\tau = 300$ μ s, which also evidences an underwater positioning of the source. However, there, it exceeds the propagation time in air τ_a , whence it follows that $L_h < L_m$ and, therefore, the source is located closer to the hydrophone than to the surface.

In individual experiments, the equality $\tau = \tau_{\min}$ was satisfied, which corresponds to a source located at the surface. This conclusion is indirectly confirmed by sound pressure amplitude measurements in air: in such cases, it multiply exceeded the theoretical estimates allowing for transient attenuation [21].

The nature of the irregular appearance of main sound packets is clarified by the cavern evolution patterns and from the fragments that detached from them, presented in Fig. 4 (frame rate 20000 fps, field of view 128×128 pixels, actual size of fragment 1.9×1.9 cm). The main focus is on visualizing the structural elements of flows, the appearance of which in time coincides with sound emission. Fast hydrodynamic processes accompanying the primary contact of the drop with the surface and emission of the shock pulse have not been analyzed here (the flow dynamics and structure at the moment of impact have been studied in [22, 23]).

The submerging drop form a cavern surrounded by the displaced crown above the fluid surface. The upper edge of the crown is irregular, and from its sharp protuberances (chevrons), chains of small drops are expelled (Fig. 4a). The bottom of the cylindrical cavern, initially flat (Fig. 4a), gradually deepens and takes a spherical shape. From the crown, circular gravitational-capillary waves extend [24].

The buoyancy and surface tension forces gradually halt the deepening of the cavern, and under their action, it begins to squeeze toward the free surface. In the drop region, a wide rosette forms, from the center of which a thick cumulative stream supported by a wide conical pedestal is expelled into the air [24].

When the stream reaches its maximum height, a secondary drop detaches from its vertex, which exceeds the primary drop in dimensions (videograms of this phase of the process are not shown). As well, the upper part of the stream assumes a conical shape. A group of short circular capillary waves propagate from the vertex of the cone over the stream that begins to

submerge. As the stream submerges, it does not necessarily retain its integrity; it can decay into several drops.

The submerging stream depresses the fluid surface and forms a conical cavern (dark triangle under the fluid surface surrounding the light cylindrical column in Fig. 4b). There are small air bubbles at the circular bottom of the cavern.

The submersion velocity of the stream base supported by the bottom of the cavern gradually decrease; when it becomes equal to zero, a very rapid collapse of the underwater part of the cavern occurs (Fig. 4c). As well, in the bulk of the fluid, the small air bubbles remain.

The bottom and sides of the cavern, which continue to contract toward the free surface, participate in complex vibrational motions. The cavern loses its regular shape, and short capillary waves begin to form on it, the amplitude of which changes in the azimuthal direction. When the head of the submerging stream reaches the uneven surface of the fluid, an irregularly shaped air cavity forms under it (Fig. 4d, center), which has vertical walls and an inclined bottom. After a certain time, a drop expelled from the vertex of the thick cumulative stream falls on the fluid surface.

The further evolution of the flow pattern evidences the existence of flows of various scales in the liquid: fast small-scale near-surface [24] and slow large-scale flows in the bulk of the fluid.

Residual flows in the region where a drop falls carry the arising cavity downward. The part of the cavern adjacent to it, which is filled with air, rapidly collapses upward by means of buoyancy and surface tension forces (Fig. 4e). Under the action of multidirectional forces, the thin bridge connecting both air cavities becomes thinner and breaks up (Fig. 4f). As a result, in the bulk of the liquid there remains an irregularly shaped air cavity which is carried downward by the background flow. The shape of the cavern changes rapidly, and the angularities are smoothed; the cavity gradually becomes spherical.

The moment of cavity detachment corresponds to the beginning of the main sound packets, which is emitted until the cavity takes an equilibrial spherical shape.

Under the cavern, there remains a group of small gas bubbles of nearly regular shape, the HF sound emission of which is lost against the intense and prolonged signal from the nonequilibrial air cavity.

The secondary drop that fell after absorption of the stream forms a subsequent cavern of conical shape; a group of gas bubbles is also adjacent to its bottom (Fig. 4g). Collapse of this cavern is accompanied by the expulsion of initially fast small drops (Fig. 4h), followed by collapse of the thin solid stream, rapidly decaying into a chain of small drops (Fig. 4i).

Figure 5 shows the phonogram of the underwater acoustic pressure, which was recorded simultaneously

with video. The high level of LF signal fluctuations reflects the effect of variable pressure resulting from displacements in the free surface. In the given experiment, synchronous input of acoustic data and video information occurs with a time error no worse than 50 μ s, which was determined by the video frame rate, 20000 fps.

The HF components of the shock pulse are shown in the extended detail (Fig. 4a). The signal character and its frequency composition is illustrated by Fig. 4b). Within the pulse time (300 μ s), the amplitude of oscillations rapidly decreases and the frequency changes irregularly in the 110–50 kHz band.

At the moment the group of small bubbles detach from the bottom of the collapsing cavern (Fig. 4c), weak sound pulse II is distinguished in the phonogram, the extended depiction of which is shown in Fig. 4c. The absence of frequency modulation can be explained by the high values of resonance frequencies of bubbles, which according to estimate [7] exceed 250 kHz; i.e., they are beyond the hydrophone band pass.

After a short quiet period, at the 187 ms, main packet III appears. Its extended image (Fig. 4d) represents a single-frequency (2.4 kHz) decreasing signal with a maximum sweep of 17 Pa. In the videogram, this event corresponds to division of the cavern into two parts, one of which (a closed irregularly shaped volume) moves into the bulk of the fluid.

After a certain time, at the 0.23 s moment, yet another sound packet, IV (sweep 30 Pa), appears in the phonogram, the extension of which is shown in Fig. 4e, and the spectrum, in Fig. 4f. This can be interpreted as an amplitude-modulated signal with a main frequency of 30 kHz. Signal IV's overall duration is significantly less than for signal III. The time of the appearance of this signal coincides with detachment of the group of small bubbles from the bottom of the conical cavern (Fig. 4g), the collapse of which generates an expulsion of individual drops and a decaying thin stream—a streamer (Figs. 4h, 4i).

Comparison of the video picture in Fig. 4 and the features of the hydrophone phonogram (Fig. 5) demonstrate that the closed irregularly shaped gas cavities, which gradually pass into equilibrial spherical volumes, serve as the main sound source in water. Closed gas cavities are formed at the moment of accelerated motion of the near-surface part of the cavern and fragments detaching from it. Namely, the main-sound-packet emission processes are timed to such moments.

The shape of the cavern changes cardinally in the evolution process. At the initial stage, the smooth walls of the cavern are gradually covered by capillary waves of rapidly changing amplitude. A simultaneous change in the depth of the cavern and the shape of the surface evidences the existence of intense flows in the near-surface layer of the fluid. The parameters of the displacement of the lower point of the cavern bottom

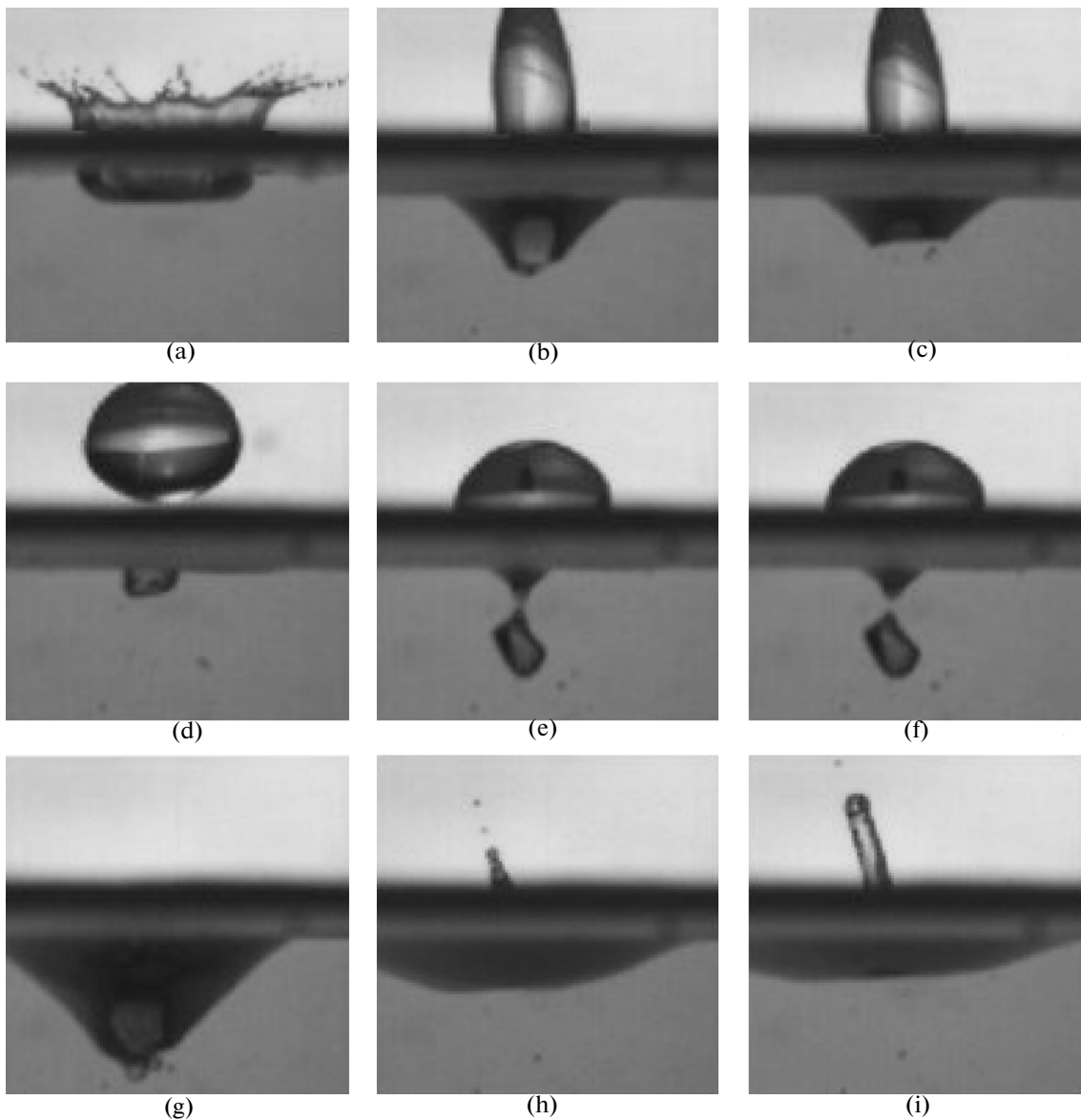


Fig. 4. Flow patterns in the drop impact region. (a–i) $t = 1.70, 162.85, 163.80, 181.30, 187.00, 187.10, 224.85, 225.15, 232.05$ ms. $H_d = 80$ cm, $U_d = 3.7$ m/s, $We = 925$, $Bo = 3.3$, $Fr = 280$, $Re = 18500$.

were chosen as the indicator of the dynamics of the processes: its depth z , vertical velocity v_z , and acceleration a .

Comparison of time dependences z , v_z , and a (curves 1–3 in the upper part of Fig. 6) with the hydrophone phonogram (Fig. 6, lower part) shows that the emission of sound packets is linked to extreme acceleration values. In the given experiment, two velocity and acceleration extrema are observed at $t = 163$ and 168 ms; two sound packets (II and III) following shock pulse I are synchronized with them.

At the initial “smooth” stage of cavern evolution, its vertical size z (counting from the level of the unperturbed surface to the lower point of the cavern) gradually increases (Fig. 6, curve 1) and then also gradually

decreases. Shock pulse I (phonogram), which is emitted only at the initial moment of contact of the drop with the surface (Fig. 6a) has an irregular character, and its level and frequency significantly change. In the decreasing area, the signal oscillates with a frequency that monotonically decreases in the range from 110 to 80 kHz (inset to Fig. 6a). Further, during a prolonged interval (up to 150 ms), there is no noticeable acoustic activity.

As in the previous experiment (Fig. 4), submersion of the cumulative stream generates a new phase in cavern evolution, the depth of which initially (curve 1) gradually increases ($t = 0.150$ – 161 ms) and then decreases. As well, the velocity and acceleration gradually change and, as follows from the videogram, the

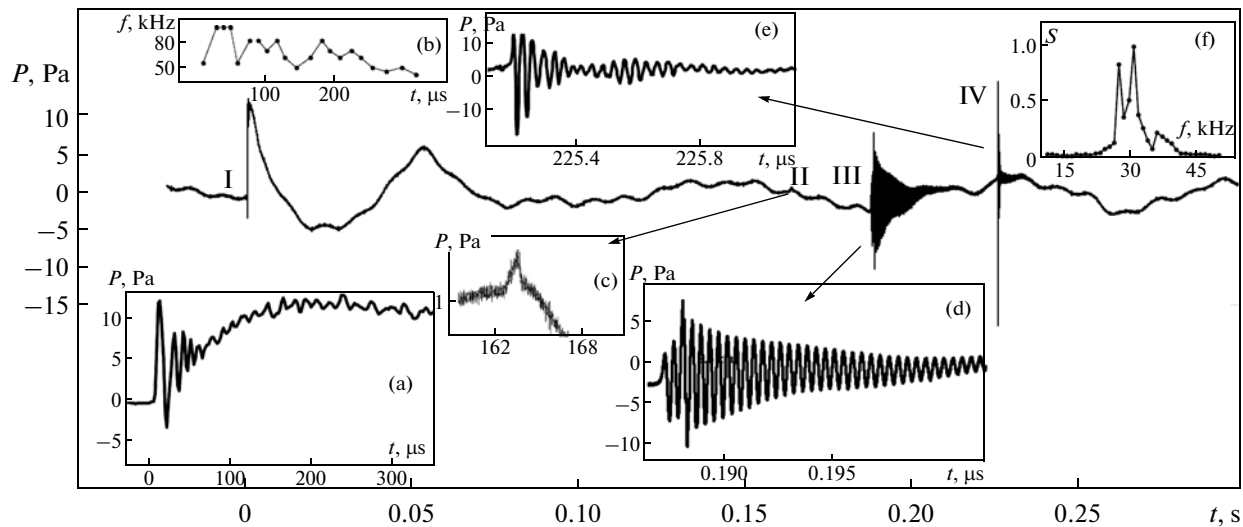


Fig. 5. Hydroacoustic signal phonogram and magnified details of sound packets upon falling of a drop from height $H_d = 80$ cm. $U_d = 3.7$ m/s, $We = 925$, $Bo = 3.3$, $Fr = 280$, $Re = 18500$.

cavern retains its integrity. At the end of this phase of the flow ($t = 168$ ms), a rapid decrease in the depth of the cavern occurs, and the velocity and acceleration reach values of 1.5 m/s and 1 km/s^2 , respectively. In the videograms, at this moment, detachment of a gas bubble from the cavern is observed, and in the hydrophone phonogram, sound packet II arises. The amplitude of the packet monotonically decreases over the course of 3 ms (Fig. 6b). In this signal spectrum, a frequency of 4 kHz dominates (inset to Fig. 6b).

The abrupt compression of the cavern is further replaced by gradual contraction to the free surface. Under the action of the secondary drop that has fallen, the cavern again begins to grow ($t = 0.177\text{--}0.191$ s), repeating in general features the evolution of the previous stage.

In the new phase of the cavern falling, at the moment of a sharp change in velocity (3 m/s) and acceleration (2 km/s^2) at $t = 197$ ms, a subsequent gas cavity detaches from the bottom. At this moment, packet III appears in the phonogram, having a duration of 3 ms (Fig. 6c) with a main frequency of 11 kHz and a less-pronounced peak at 4.5 kHz (inset to Fig. 6c). Like in the previous experiment, sound emission occurs at moments of a sharp change in the shape and area of the cavern surface and detachment of gas cavities of nonequilibrium shape. As well, extreme accelerations are recorded, the source of which is pressure gradients arising as a result of the decrease in surface energy of the cavern.

The variability of the sound emission process even under weak changes in the experimental conditions is illustrated by the following example in which a relatively small increase (0.4 m/s) in the drop velocity led to a different scenario for the development of the process (Fig. 7).

Here, the parameter behavior of the motion of the cavern bottom in its main features is analogous to the previous experiment. At the moment the drop comes in contact with the surface, shock pulse I (phonogram) arises, which has an irregular shape (Fig. 7a) and a wide spectrum (inset to Fig. 7a) in the 100–20 kHz range.

The depth of the cavern generated by the drop impact at first gradually increases and decreases ($t = 0\text{--}50$ ms) and then stabilizes ($t = 50\text{--}145$ ms), after which a new increase–decrease phase follows ($t = 145\text{--}165$ ms), at the end of which a relatively small jump is observed in the z , v_z , and a values. Pulsating sound packet II with a duration of 5 ms pertains to this moment (Fig. 7b). In the spectrum, two peaks are pronounced at close frequencies (10.1 and 10.6 kHz), as well as side lobes in the range from 9.0 to 12.6 kHz (Fig. 7c).

Further, the depth of the cavern generated by submersion of the cumulative stream increases weakly (to the moment $t = 178$ ms), following which it falls to zero (the fluid surface evens out). At this moment, a closed air cavity detaches from the bottom of the cavern, and at the hydrophone output, pulsating sound packet III appears with a duration of 15 ms (Fig. 7d). There is a main frequency of 3.3 kHz in its spectrum (Fig. 7e), as well as several side frequencies in the range of 2.5–4.5 kHz.

Then, yet another deepening phase of the cavern follows, caused by the falling of the secondary drop that detached from the cumulative stream ($t = 180\text{--}240$ ms). At the moment of the sharp change in acceleration of the bottom, detachment of gas bubbles occurs, which is accompanied by the appearance at the hydrophone of sound packet IV (Fig. 7f). This packet is uniform in frequency composition (main fre-

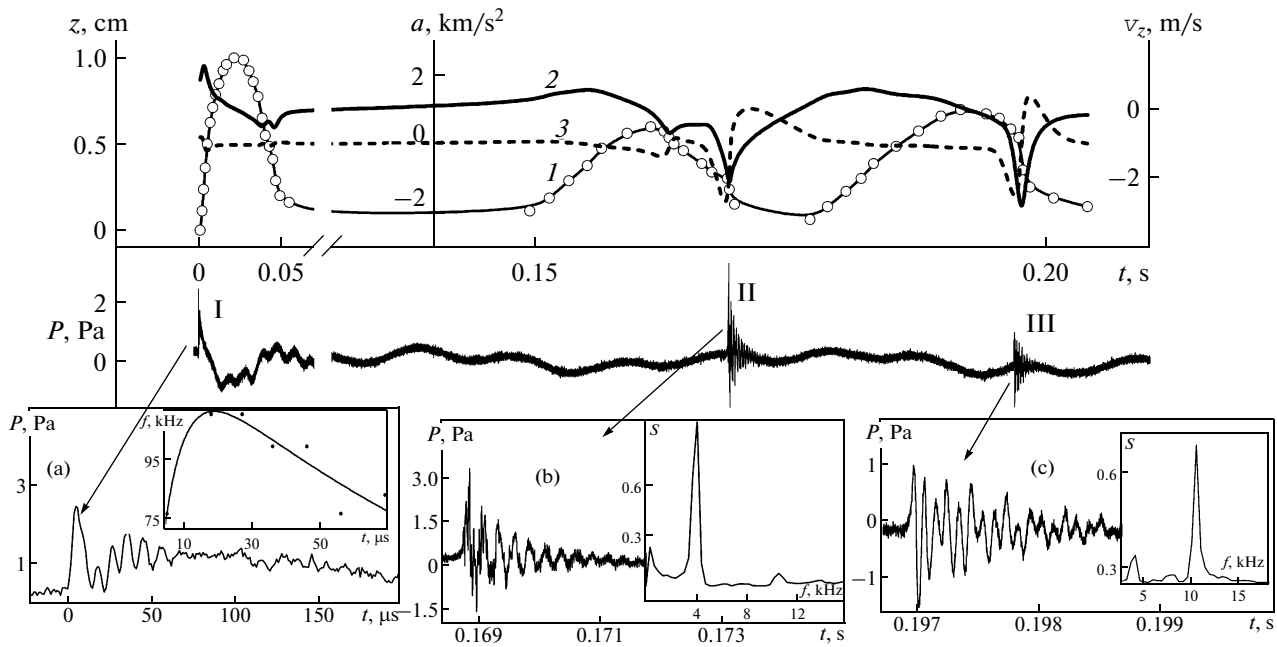


Fig. 6. Depth z (1), vertical velocity v_z (2), and acceleration a (3) of the lower edge of the cavern and phonogram of underwater sound pressure P . $H_d = 50$ cm, $U_d = 2.9$ m/s. $We = 570$, $Bo = 3.3$, $Fr = 170$, $Re = 14500$. Recording speed 10000 fps.

quency 3.5 kHz (Fig. 7g), and it attenuates nearly monotonically.

Mutual analysis of video frames, phonograms, and cavern evolution curves make it possible to define the main moments relating the hydrodynamic processes to sound emission. So, sound packet III in Fig. 5. appears at the moment of 178 ms, when an irregularly shaped air cavity detaches from the cavern (Fig. 4d).

The cavity oscillates, constantly changing form, and in the end result transforms to a sphere with a diameter of $d_c = 0.23$ cm. At that, the gas cavity volume hardly changes at all. Packet III emits for 20 ms—precisely the same time required by the detaching cavity for transition to spherical shape. The oscillation frequency is 2.5 kHz, which is close to the calculated value of $f = 656/d_c$ [7].

Sound emission of a resonating cavity is not necessarily accompanied by its detachment from the main cavern. So, the beginning of packet IV, the last in the series, coincides with the fast displacement of two small bubbles (Fig. 4h) carried away by the bottom of the cavern at the peak acceleration. The size of bubbles is almost 0.02 cm, which corresponds to the resonance frequency of 33 kHz, located near the spectral peaks of packet IV (Fig. 5f).

In a similar scenario, hydrodynamic and acoustic processes proceed in the other experiments as well. The main differences are in the dimensions of the closed gas cavities, initially irregular in shape, participating in accelerated movement, as a result of which similar parameters in the emitted acoustic packets differ: amplitude, modulation frequency, duration.

We note also that the initial amplitudes of the main packets, as a rule, do not go beyond 10 Pa, but much higher levels are also encountered (e.g., packet IV, Fig. 5), comparable to those observed under artificial excitation of bubbles by powerful ultrasound signals with an amplitude on the order of 10^5 Pa [25].

The conducted experiments make it possible to formulate the following physical model for sound generation by a droplet falling on the surface of a fluid.

A falling drop transfers matter, pulse, and energy. Upon collision and subsequent oscillations, the area of the free surface of the fluid changes and these processes are characterized by two characteristic timescales: fast (confluence process) and slow (capillary vibrations, cavern movement). Fluid dynamics elements are associated with each scale. Fast confluence processes form fine structure high-speed flows in the near-surface layer. Slow processes form large-scale circulation flows in the bulk of the fluid. This slowly attenuating flow plays the main role in cavern deformation, the formation of closed gas cavities in the bulk of the liquid, and their being carried away from the free surface.

The initial (shock) sound pulse forms at the moment of contact of the lower part of the drop with the concave surface of the receiving fluid. The caving in of the free surface occurs as a result of the action of advancing perturbations, creating an increase in pressure in front of the moving high-drag body and air displacement from the closing gap between the drop and the receiving fluid. As well, nearly simultaneously over the entire surface area of the closing, “destruction” of

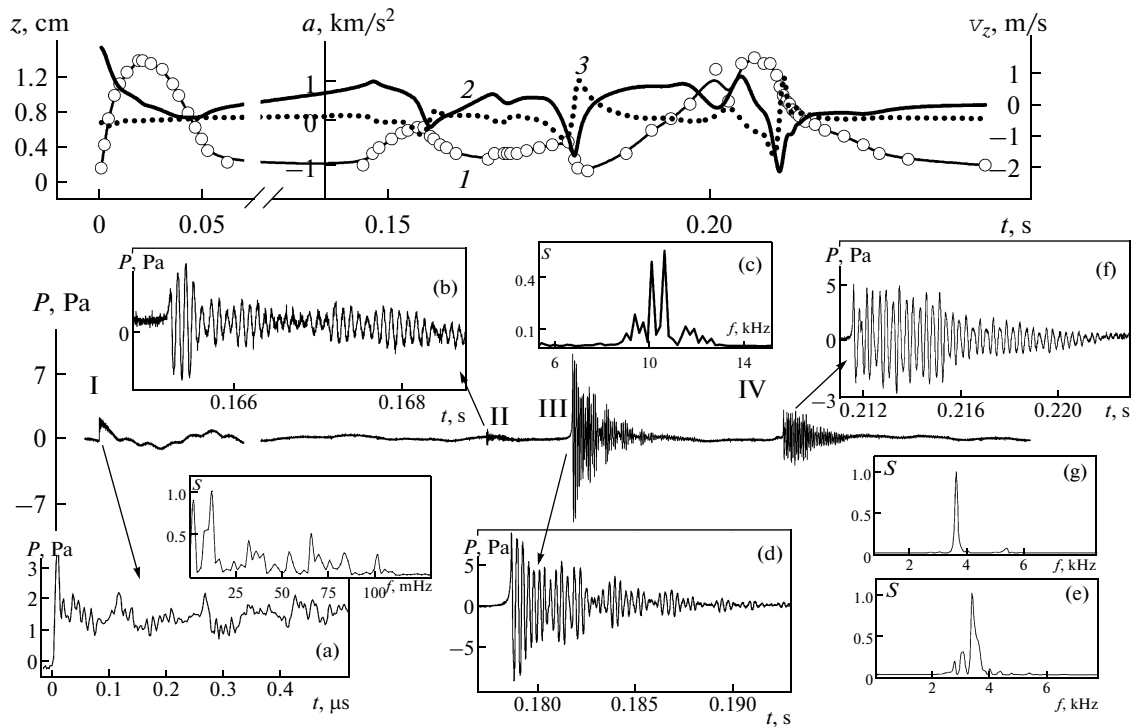


Fig. 7. Depth z (1), vertical velocity v_z (2), and acceleration a (3) of the lower edge of the cavern and phonogram of sound pressure P . $H_d = 72$ cm, $U_d = 3.3$ m/s, $We = 735$, $Bo = 3.3$, $Fr = 222$, $Re = 16500$. Recording speed 2000 fps.

the free surface of the fluid, release of surface energy, and the formation of a pressure pulse with a steep leading edge occur. The contact boundary between the remainder of the drop and the cavern in the receiving fluid at the initial stage moves at supersonic speed, creating perturbations in the shape of the surface, leading to the trapping of small gas bubbles [22, 23].

The main sound packets are emitted at a later stage, when there are differently scaled flows in the receiving fluid: a circulation flow related to the action of the drop pulse and a small-scale HF flow as a result of the changing area of the free surface. Small-scale processes lead to irregular deformation of the free surface and the formation of gas cavities of arbitrary form (nonequilibrium, with sharp boundaries and conical apices), which are pulled by large-scale flows into the bulk of the fluid.

The forming air cavities resonate under abrupt force action, an indicator of which is extreme accelerations arising during displacement of the bottom of the cavern and detachment of closed gas cavities. The frequency of emitted sound is determined by the equilibrium size of the forming gas cavity, and the duration of sounding, by the transition time of the initially irregularly shaped cavity into a state of equilibrium.

ACKNOWLEDGMENTS

The study was financially supported in part by the Russian Federation Ministry of Education and Science (Federal Target Program "Research and Development in Priority Directions of Russia's Scientific-Technological Complex for 2007–2012," State Contract no. 16.518.11.7059) and the Russian Academy of Sciences (RAS Presidium Program no. 20 "Fundamental Problems of Oceanology").

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